

Laxmi Charitable Trust's
Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce
Department of Information Technology (B.Sc.I.T Semester V)
Applied business analytics

Practical – I

Roll No.:T039	Name:Bhavesh Ramakant Salaskar
Class:TYIT	Batch:1
Date of Assignment:24-07-2026	Date/Time of Submission:24-07-2026

Python

```
# Import Required Libraries
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
```

Python

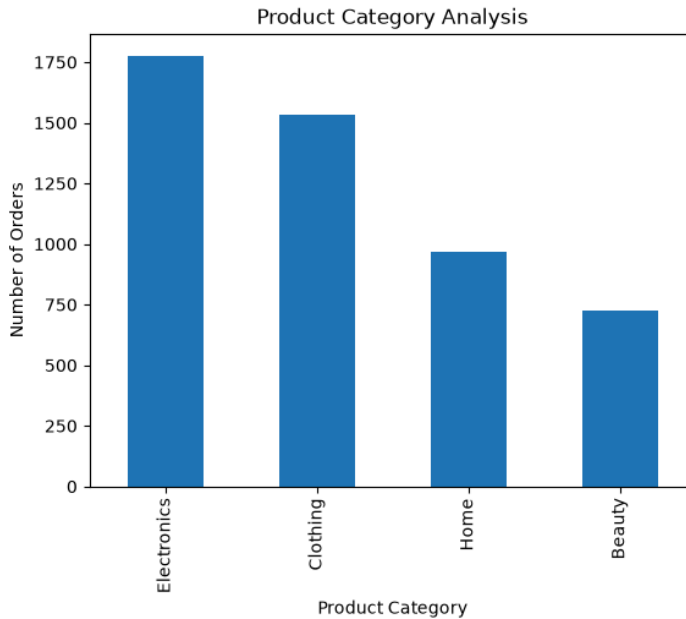
```
# Load Dataset
df = pd.read_csv("ecomer.csv")
```

Analysis 1 - Product category analysis

Python

```
print(df["product_category"].value_counts())
df["product_category"].value_counts().plot(kind="bar")
plt.title("Product Category Analysis")
plt.xlabel("Product Category")
plt.ylabel("Number of Orders")
plt.show()
```

```
product_category
Electronics    1777
Clothing       1531
Home           969
Beauty         723
Name: count, dtype: int64
```

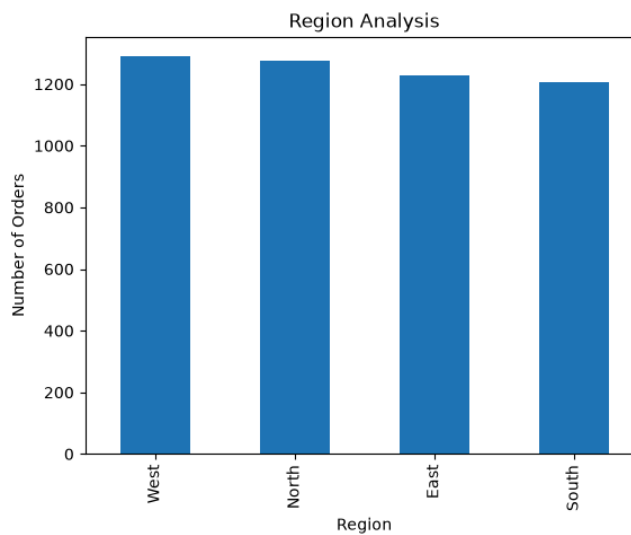


Analysis 2 - Region Analysis

Python

```
print(df["region"].value_counts())
df["region"].value_counts().plot(kind="bar")
plt.title("Region Analysis")
plt.xlabel("Region")
plt.ylabel("Number of Orders")
plt.show()
```

```
region
West    1289
North   1276
East    1227
South   1208
Name: count, dtype: int64
```

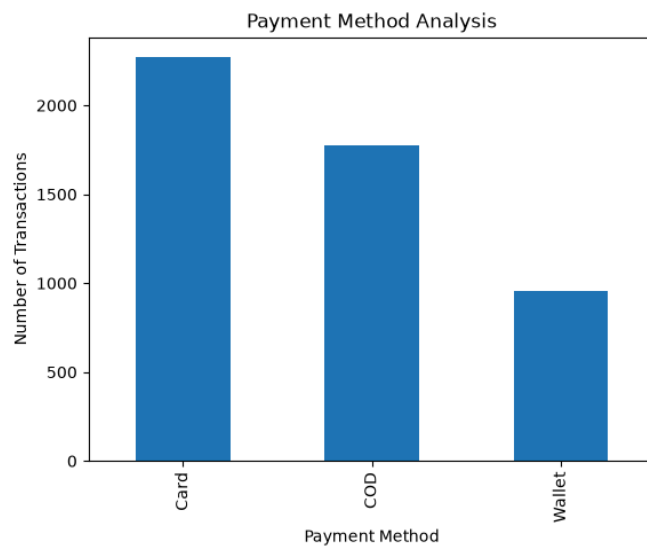


Analysis 3 - Payment Method Analysis

Python

```
print(df["payment_method"].value_counts())
df["payment_method"].value_counts().plot(kind="bar")
plt.title("Payment Method Analysis")
plt.xlabel("Payment Method")
plt.ylabel("Number of Transactions")
plt.show()
```

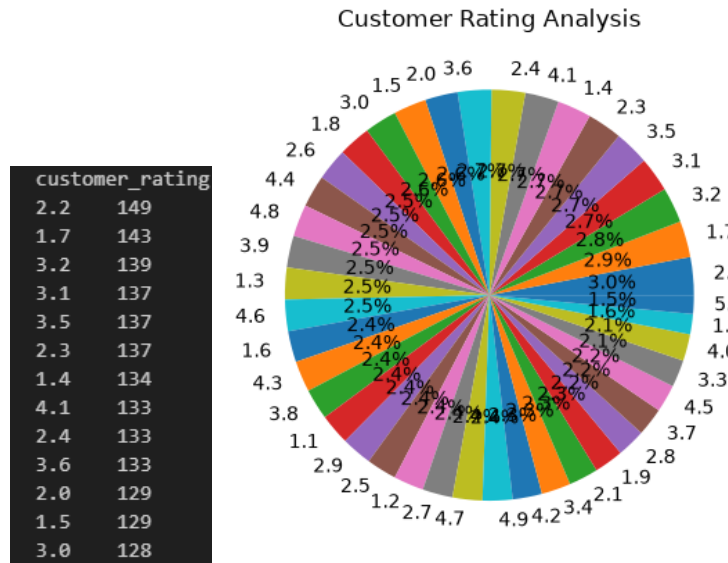
```
payment_method
Card      2270
COD       1774
Wallet     956
Name: count, dtype: int64
```



Analysis 4 - Customer Rating Analysis

Python

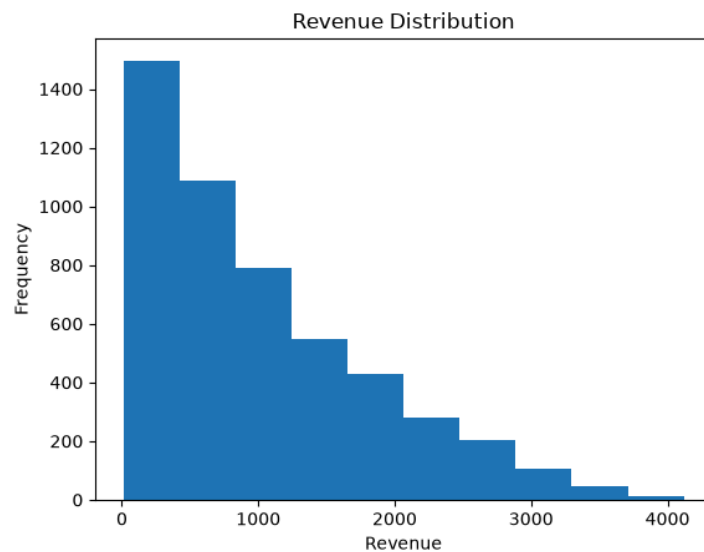
```
print(df["customer_rating"].value_counts())
df["customer_rating"].value_counts().plot(kind="pie",
autopct="%1.1f%%")
plt.title("Customer Rating Analysis")
plt.ylabel("")
plt.show()
```



Analysis 5 - Revenue Analysis

Python

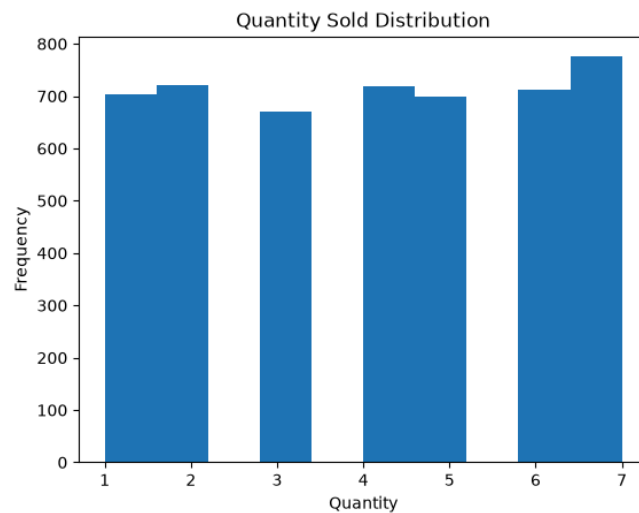
```
plt.hist(df["revenue"], bins=10)
plt.title("Revenue Distribution")
plt.xlabel("Revenue")
plt.ylabel("Frequency")
plt.show()
```



Analysis 6 - Quantity Sold Analysis

Python

```
plt.hist(df["quantity"], bins=10)  
plt.title("Quantity Sold Distribution")  
plt.xlabel("Quantity")  
plt.ylabel("Frequency")  
plt.show()
```



Analysis 7 - Unit Price vs Revenue

Python

```
plt.scatter(df["unit_price"], df["revenue"])  
plt.title("Unit Price vs Revenue")  
plt.xlabel("Unit Price")  
plt.ylabel("Revenue")  
plt.show()
```



Analysis 8 - Correlation Analysis

Python

```
sns.heatmap(df.corr(numeric_only=True), annot=True, cmap="Oranges")  
plt.title("Correlation Heatmap")  
plt.show()
```

